Evit John, BDS, MPH, PHD Candidate

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EDUCATION

Doctor of Philosophy (Ph.D.) - Biostatistics 2021 - Present
School of Public Health
University of Texas Health Science Center at Houston (UTHealth)
Houston, Texas

Master of Public Health (MPH) 2018 - 2020
School of Public Health
University of Texas Health Science Center at Houston (UTHealth)
Houston, Texas

Bachelor of Dental Sciences (BDS) 2011 - 2016
Manipal College of Dental Sciences (MCODS), Manipal University
Manipal, India

EXPERIENCE

Graduate Research Assistant May 2022 - Present
Van Loo Lab
Department of Genetics, MD Anderson Cancer Center, Houston, Texas

- Develop and optimize computational pipelines in R, Python, and Snakemake for analysis of bulk and single-cell DNA sequencing and RNA-seq data across multiple cancer types, including ovarian, oral, renal, lung, and biliary cancers
- Pioneering a novel single-cell copy number variant (CNV) detection method to resolve intratumoral heterogeneity and inform models of cancer evolution and clonal dynamics
- Integrate multi-omic datasets to characterize how molecular alterations drive cancer progression and metastasis, linking genomic features to clinical phenotypes and patient outcomes
- Apply statistical and computational frameworks to clinical risk stratification using tumor genomic profiles, translating biological findings into clinically actionable insights
- Collaborate within a multidisciplinary research team while independently managing concurrent analysis workflows across diverse cancer cohorts

Teaching Assistant May 2022 – August 2022
PH1710 - Intermediate Biostatistics, Department of Biostatistics
University of Texas Health Science Center - School of Public Health, Houston, Texas

- Observed and evaluated student performances, and development of 35 students.
- Evaluated and revised lesson plans and course content to facilitate student-centered learning.
- Enhanced student learning by optimizing teaching techniques based on individual learning styles during TA sessions.
- Graded and tracked homework, tests, midterms, and final exams.
- Assumed responsibility for the classroom during the head teacher's absence.

Research Data Coordinator November 2020 – May 2022
Department of Surgical Oncology
MD Anderson Cancer Centre, Houston, Texas

- Curated and maintained the Medullary Thyroid Carcinoma Registry, managing end-to-end data abstraction from clinical systems (EPIC, REDCap) and ensuring data integrity across patient records for longitudinal research use

- Designed and executed epidemiological investigations using registry data to support drug research and clinical decision-making, producing routine analytical reports for stakeholder review
- Translated complex clinical and genomic registry data into structured datasets, enabling downstream statistical analysis and population-level insights into Medullary Thyroid Cancer
- Communicated research findings through presentations to cross-functional audiences including clinicians, researchers, and patients, demonstrating strong scientific communication skills
- Trained and mentored graduate students in clinical data management best practices, supporting team-wide research productivity

Research Intern – Practicum Volunteer March 2020 – August 2020
Oral Health Program
Texas Health Institute, Austin, Texas
Mentor: Dr. Josephine Wolfe, Ph.D., RDH, CHES

- Plans, designs, conducts and coordinates epidemiological investigations for research from the registry.
- Designed and implemented a survey of oral health care providers to assess their knowledge, attitudes, and beliefs about the HPV vaccine.
- Formulated an intervention strategy and communication and education packet for oral health care providers and parents.

Graduate Research Assistant January 2019 – August 2020
Department of Surgical Oncology
MD Anderson Cancer Centre, Houston, Texas

- Provided support for the completion of and updating of the Medullary Thyroid Carcinoma Registry
- Managed data abstraction from EPIC, RedCap entry, and patient file and data management for further modification and arrangement for the registry.

Lecturer, Department of Oral and Maxillofacial Surgery May 2017 – August 2018
Manipal College of Dental Sciences
Manipal University, Manipal, India

- Performed routine dental extraction, and OPD-based jawbone surgery.
- Assisted in the operation theater on major cases such as orthognathic surgery, head and neck oncology, facial trauma, cleft lip, and palate, etc.
- Instructed and guided students in the clinical setting and guided basic extractions, and jaw surgeries
- Supervised three small-scale student projects
- Created educational videos on 'Medical Emergencies in Dental Practice' and 'Suturing techniques.
- Created a new questionnaire-based protocol for clinicians to assess new patients with Temporomandibular Joint Disorders

GRANTS

Current:

- Head and neck premalignant and malignant tumors – UTMDACC - Precancer Atlas Head & Neck Initiative
- Multiregional Cholangiocarcinoma - Department of Defense RCRP Idea Award & NCI-R21

Past:

- A novel natural history study of medullary thyroid carcinoma: Incorporating the patient perspective to inform advanced disease management – NIH R01

PUBLICATIONS

Manuscripts in preparation

John E.R., et al. "Genomic and spatial transcriptomic heterogeneity of multi-region cholangiocarcinoma." – Journal of Hepatology

Morris B., **John E.R.,** Cheng Y., et al. "Tracking tumor evolution under treatment" – Cancer Research Review

Tarabichi M, Lesluyes T., et al. "A generic framework for DNA copy number quantification with ASCAT"

Published

John, Evit, et al. "Reconstructing oral cavity tumor evolution through brush biopsy." *Scientific Reports* 14.1 (2024): 22591.

John ER, Lesluyes T, Baker TM, Tarabichi M, Gillenwater A, Wang JR, Van Loo P and Zhao X. 2023. Reconstructing oral cavity tumor evolution through brush biopsy. bioRxiv doi: 10.1101/2023.11.29.569216 (preprint)

Adzrago, D., Obekpa, E. O., Suragh, T. A., **John, E. R.,** Yeh, P. G., Gallardo, K. R., & Wilkerson, J. M. (2022). Kratom use categories and their associations with co-occurring substance use and mental health disorder symptoms during the COVID-19 pandemic. *Drug and alcohol dependence*, *239*, 109605.

Kudva, A., Kumar, M., **John, E. R**., & Dhara, V. (2022). Occurrence of Second Oral Potentially Malignant Disorder following Excision of Primary Lesion: A Prospective Study of Cases from a Tertiary Care Centre. *Journal of Maxillofacial and Oral Surgery*, 1-6.

Kapur, N., Singla, N., Kudva, A., & **John, E. R.** (2021). Evaluation of quality of life in patients surgically treated for potentially malignant oral lesions. *Indian Journal of Cancer*, *58*(3), 371-377.

Smriti, K., **John, E**., Pentapati, K. C., Gadicherla, S., & Bhagania, M. (2020). Intra-lesional medicaments for the management of intra-osseous lesions of maxilla and mandible-systematic review. *Journal of International Society of Preventive & Community Dentistry*, *10*(1), 36.

Nivedita, S., **John, E. R.,** Acharya, S., & D'costa, V. G. (2019). Prophylactic extraction of non-impacted third molars: is it necessary?. *Minerva Stomatologica*, *68*(6), 297-302.

John, E. R., Kadaikal, B., Manocha, S., Rao, S., & Ballal, N. V. (2018). Evaluation of micro crack propagation in root canal dentin after the use of Reciproc and Waveone Gold Rotary systems. *Int J Clin Dent*, *11*, 201-8.

Gadicherla, S., Pentapati, K. C., Singh, A., **John, E. R.,** & Smriti, K. (2018). Evaluation of a new ergonomic position for the operator/clinicians for the extraction of mandibular right posterior molar teeth. *Journal of International Oral Health*, *10*(1), 36.

Srikanth, G., **John, E. R.**, & Pentapati, K. C. (2017). Oral and Maxillofacial approach for the treatment of Obstructive Sleep Apnea-Review. *Journal of Pharmaceutical Sciences and Research*, *9*(6), 840.

SERVICE TO PROFESSION:

Papers reviewed:
Cell reports medicine: A changed perspective on the genetic progression of cancer by longitudinal analysis of oral lesions; Wils LJ et al.
Journal of Pathology: Genetic progression of cancer; JB Poell et al.

LEADERSHIP EXPERIENCE

- UTHealth Student Inter-council 2018 – 2020
  - Historian:
  - School of Public Health Representative
  - University Academic Council
  - UT System Student Advisory Council
- President, MCODS Student Council 2015 – 2016
- General Secretary, Manipal International Network of Education (MINE) 2015 –2016
- National Officer of Medical Education, Medical Students Association of India (MSA-I) 2013 – 2014
- Student Representative, MCODS Student Council 2011 –2012

PROVISIONAL PATENTS

- Specialized electronic toothbrushes for patients with cerebral palsy and other motor dysfunctions (Provisional patent no – 201841016406).
- Specialized device for debridement of pericoronal flap

AWARDS AND HONORS

- Delta Omega Honor Society – Delta Omega Chapter – School of Public Health, UTHealth Science Center, Houston, 2020.
- Dr. David W. Martin Memorial Scholarship, UTHealth Science Centre – School of Public Health
- Schull Institute Summer Scholar, 2019.
- Provenance, Manipal University Technology Business Incubator (MUTBI) Business B Plan Competition, 2016.
- Best Poster Award, 1st International Conclave of Health Technical Research, 2015.
- Student Leadership Program, International Dental Exhibition, and Meet, 2015.
- Entrepreneurship Awareness Camp, Manipal Academy of Higher Education, 2015.
- Keynote Speaker, International Youth Summit, Yuva Varsha, 2013.
- Leadership Workshop for Student Leaders of Manipal, Manipal Academy of Higher Education, 2012
- Leadership Training Camp, M.R. Pai Foundation, 2011.